

rinode: Constant-Time Zero-Copy File Preservation and Metadata-Intact Restoration on Linux Virtual File System

A Technical Whitepaper on Inode Lifecycle Mechanics, Mount-Local Vaulting, and VFS Semantics

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Abstract

In standard Linux environments, unlinking a file through the POSIX `unlink(2)` interface immediately severs its directory entry binding and decrements the inode's hardlink reference count. When this count reaches zero and all active file descriptors close, filesystems like ext4, XFS, and Btrfs purge extent trees, mark block group allocation bitmaps as free, and clear inode metadata records. Conventional user-space mitigations, such as desktop trash specifications (FreeDesktop.org Trash), prevent immediate data destruction by moving files across mount points via deep byte-by-byte copies. This introduces severe $O(N)$ write amplification, depletes I/O bandwidth, delays execution when removing large file hierarchies, and discards lower-level inode provenance.

This paper presents the architectural design, kernel interfaces, and formal evaluation of `rinode`, a high-performance systems utility engineered for Linux that achieves constant-time $O(1)$ deleted file preservation without duplicating data blocks. By coupling mount-boundary discovery via `statx(2)` with `STATX_MNT_ID`, atomic same-filesystem preservation through `renameat2(2)`, Write-Ahead Logged (WAL) SQLite metadata indexing, and dual-mode restoration supporting copy-on-write reflink clones (`ioctl(FICLONE)`), `rinode` preserves original inode numbers, permissions, timestamps, and extended attributes with zero data duplication. We analyze the VFS invariants governing directory entries and inodes, evaluate transaction safety under concurrent modifications, examine race mitigation via directory file descriptors, and establish empirical benchmarks demonstrating sub-millisecond preservation of multi-gigabyte datasets and deep directory trees.

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1 Problem Statement & Motivation

1.1 The Linux Inode Lifecycle and the Unlink Problem

In UNIX-derived operating systems adhering to POSIX standards, the lifecycle of persistent storage entities is mediated through two distinct abstraction layers:

1. **Directory Entries (Dentries):** Name-to-inode mappings stored within directory data structures, forming the human-navigable filesystem hierarchy.
2. **Index Nodes (Inodes):** On-disk metadata structures tracking file attributes (ownership, mode, timestamps, access control lists) and the block pointer mapping (extent trees or indirect block arrays) referencing raw data blocks on storage media.

When an application or shell invokes `rm(1)`, the underlying command executes the `unlink(2)` or `unlinkat(2)` system call. Crucially, the Linux kernel does not possess a native “delete” operation. Instead, `sys_unlink` decrements the target inode’s link counter (`i_links_count`). If `i_links_count` drops to zero, the inode transitions into an orphaned state. The moment every open file description referencing that inode is closed by active processes (i.e., when `f_count` in `struct file` reaches zero), the Virtual File System (VFS) invokes the filesystem-specific `evict_inode` callback.

1.2 Extent Tree Deallocation and Inode Recycling

In modern journaling Linux filesystems such as `ext4`, the invocation of `ext4_evict_inode` triggers an immediate and irreversible teardown:

- **Extent Tree Truncation:** The `ext4` extent tree root, stored within the inode’s `i_block` field, along with any intermediate index nodes and leaf extents, is traversed. All referenced block clusters are returned to the block group’s allocation bitmap.
- **Bitmap Invalidation:** The kernel modifies the block bitmap block in memory and commits the deallocation to the Journaling Block Device (JBD2).
- **Inode Zeroing:** The inode table entry is cleared, and the inode allocation bitmap is updated to mark the inode number as free for immediate reallocation.

Once an extent tree is severed and the journal transaction commits, reconstructing fragmented or large files requires brute-force block scraping using heuristic tools (e.g., `photorec`, `ext4magic`). On active multi-user or heavily loaded systems, freed blocks are often overwritten within milliseconds by subsequent write operations, rendering post-mortem recovery impossible.

1.3 Taxonomy of Existing Mitigations

To mitigate catastrophic unlinks, existing Linux tooling employs three paradigms, each exhibiting critical limitations:

1. **FreeDesktop.org Trash Specification:** Standard desktop environments (GNOME, KDE) maintain trash folders under `$XDG_DATA_HOME/Trash`. When removing a file situated on a different mount point, the trash engine is forced to copy the entire file payload byte-by-byte before issuing an unlink. Deleting a 100 GB virtual machine disk image or an extensive container rootfs consumes hundreds of seconds of disk I/O and exhausts storage. Furthermore, ownership, inode numbers, and original extended attributes are frequently distorted.
2. **Filesystem-Level Snapshots:** Copy-on-Write (CoW) systems like `Btrfs` and `ZFS` allow subvolume snapshotting. However, snapshots operate at subvolume or filesystem granularity rather than targeting individual transactional file unlinks. They require pre-allocated CoW capacity, cannot run on conventional `ext4/XFS` root partitions, and do not track user intent at the CLI boundary.

Table 1: Comparison of Linux Deleted File Recovery and Retention Approaches

Mechanism	Time Complexity	Metadata Intact	VFS Scope	I/O Overhead
Standard <code>rm</code>	$O(\text{extents})$	No (Purged)	Universal	Minimal (Bitmap update)
FreeDesktop Trash	$O(N)$ cross-mount	Partial (Path/Date)	Desktop-only	Extreme (Byte copying)
Btrfs/ZFS Snapshots	$O(1)$ subvolume	Yes	CoW-only	Periodic write amp
Raw Disk Carving	$O(\text{disk size})$	No (Heuristic)	Extent-dependent	Extreme read scan
rinode	$O(1)$	Yes (Exact Inode)	Universal VFS	Zero-Copy Meta-data

3. **Block Carving and Scavenging:** Tools analyzing unallocated clusters lack directory hierarchy context, lose original filenames, fail completely on non-contiguous extents, and require administrative access to raw block devices.

1.4 The Zero-Copy Preservation Hypothesis

The central premise of `rinode` is that **an inode's link counter never needs to drop to zero upon user deletion**. By atomically redirecting the directory entry from its parent hierarchy into a secure, permissions-locked vault residing within the exact same filesystem boundary, the inode retains its exact physical data extents, ownership, timestamps, and permissions without issuing a single block write.

2 Linux Virtual File System (VFS) Fundamentals

2.1 Directory Entries (Dentries) vs. Inodes

The Linux VFS decouples file nomenclature from file identity. An inode is indexed by a filesystem-wide 64-bit unsigned integer (`ino_t`) identifying an entry in the filesystem's on-disk inode table. Directories are specialized files containing tables of directory entries (`struct dentry` in the kernel VFS dcache), each pairing a UTF-8 character string with a target inode number.

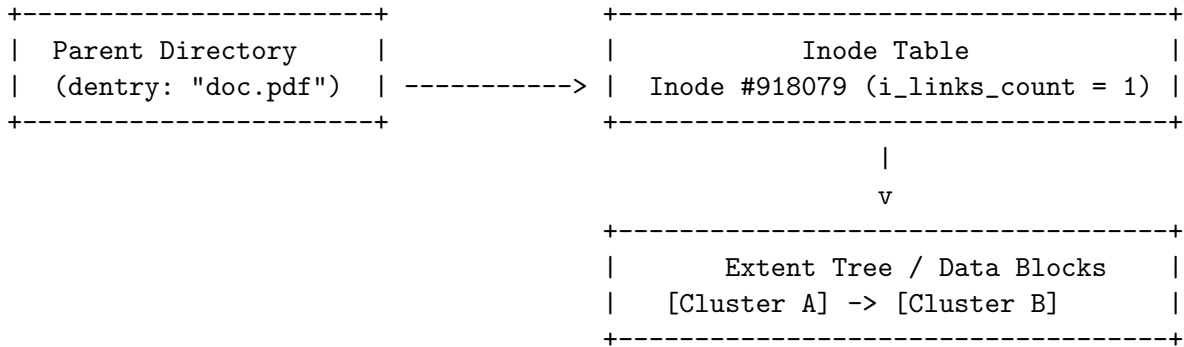


Figure 1: Decoupled architecture of VFS directory entries and underlying index nodes.

Multiple dentries may map to a single inode (hard links). An inode is eligible for reclamation if and only if:

$$i_links_count = 0 \quad \wedge \quad \sum_{\text{processes}} f_count(\text{inode}) = 0 \quad (1)$$

2.2 Link Count Semantics and the Atomic Rename Guarantee

Traditional utility designs attempting undelete features often relied on issuing `link(target, vault)` followed by `unlink(target)`. This two-step pattern violates multiple POSIX and security guarantees:

1. **Non-Atomicity:** A process failure or power interruption between `link` and `unlink` produces persistent duplicate dentries.
2. **Directory Incompatibility:** POSIX explicitly forbids creating hard links to directories (EISDIR) to prevent cycle formation in filesystem directed graphs.
3. **Kernel Hardlink Restrictions:** Modern Linux kernels enforce `fs.protected_hardlinks=1` (via `sysctl`), forbidding unprivileged users from hardlinking to files they do not own unless they already possess read/write access.

In contrast, the `rename()` and `renameat2()` system calls guarantee atomic transition. In the VFS layer, `vfs_rename` locks the parent directory inodes, creates the new dentry binding, severs the old binding, and releases locks in a single synchronized transaction. The inode's `i_links_count` remains strictly unchanged ($n \rightarrow n$), bypassing all directory and hardlink protection barriers.

2.3 Hardlinks vs. Reflink Clones

Understanding the distinction between hard links and Copy-on-Write reflinks is crucial for restoration semantics:

- **Hardlink:** A secondary dentry referencing the *identical* inode number. Any subsequent write via either dentry modifies the shared physical blocks and alters the inode's modification timestamp (`mtime`) across both references.
- **Reflink** (`ioctl(FICLONE)`): Introduced in Linux 4.5 and supported by Btrfs and XFS, a reflink creates a *completely independent inode* pointing to the *identical shared physical extents*. Neither inode shares metadata or write operations. When either inode is modified, the underlying CoW engine allocates new physical blocks only for the mutated sectors, preserving the unmodified snapshot.

2.4 Mount Identifiers, Subvolumes, and the EXDEV Barrier

The VFS imposes a hard structural invariant: directory operations cannot cross filesystem or mount boundaries. If a caller attempts to `rename(old, new)` where `old` and `new` reside on distinct superblock instances or distinct VFS mount points, the kernel immediately halts with error code `EXDEV` ("Invalid cross-device link"):

```
1 if (old_path.mnt != new_path.mnt)
2     return -EXDEV;
```

Listing 1: Kernel mount boundary check in `fs/namei.c`

On systems utilizing Btrfs or advanced container runtimes, subvolumes and bind mounts frequently share the identical `st_dev` device identifier while existing as distinct mount structures within the kernel's internal mount tree. Traditional `stat(2)` returns identical `st_dev` values for two subvolumes on the same storage pool, obscuring the fact that a `rename()` across them will trigger `EXDEV`.

To navigate this limitation, `rinode` leverages Linux 5.8+ `statx(2)` with the `STATX_MNT_ID` mask. The resulting `stx_mnt_id` field exposes the kernel's internal 64-bit mount identifier, providing deterministic discovery of mount and subvolume boundaries.

3 Architecture & System Design

3.1 System Overview

rinode is implemented as a native systems binary in Rust, with zero external runtime dependencies. It consists of six interacting subsystems:

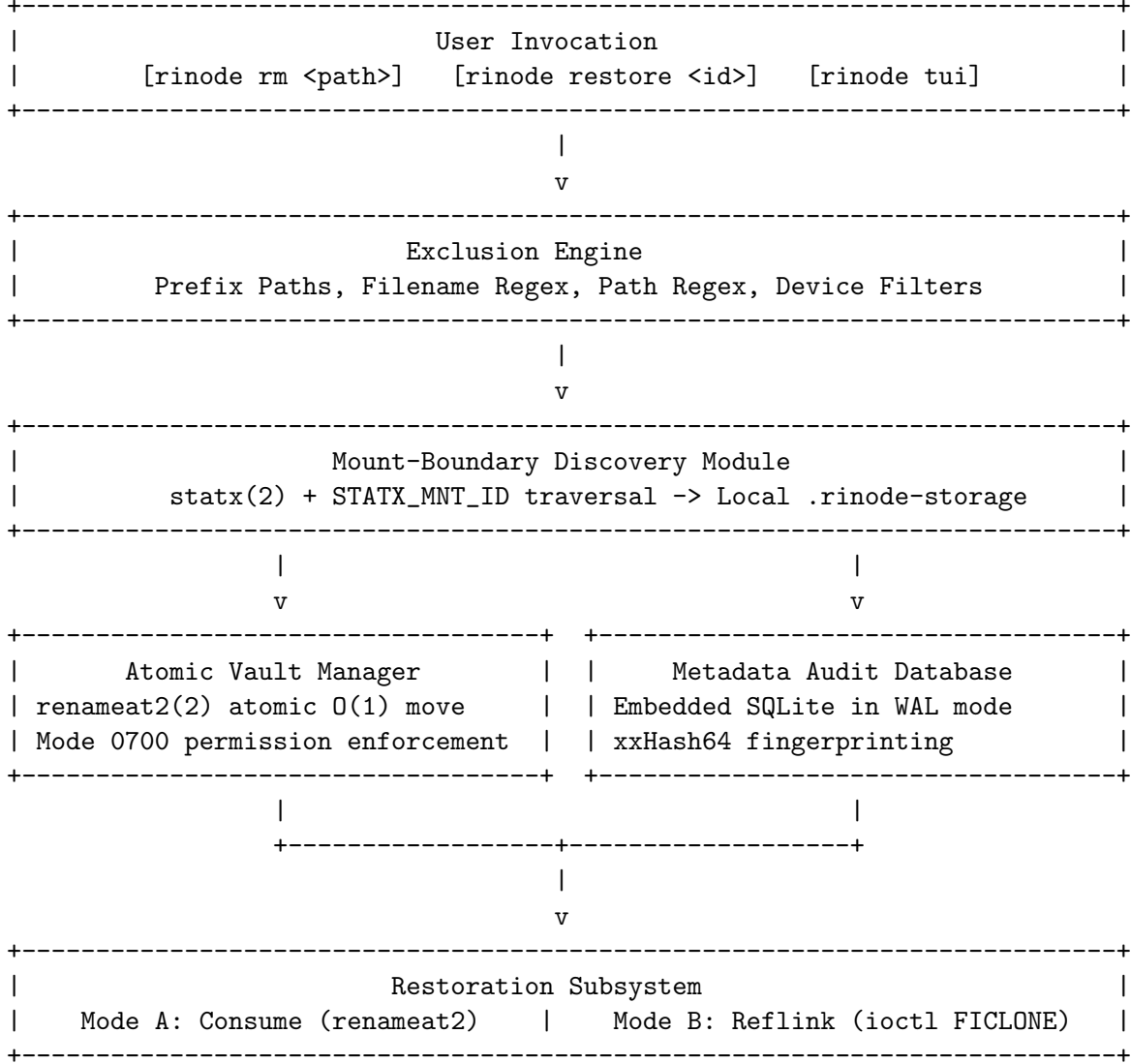


Figure 2: System architecture of rinode showing transactional component flow.

3.2 Per-Mount and Subvolume Storage Topology

To guarantee that file relocations never trigger EXDEV, rinode resolves the mount topology dynamically. When invoked against a target path:

1. The utility executes `statx(AT_FDCWD, target, AT_SYMLINK_NOFOLLOW, STATX_ALL | STATX_MNT_ID)`.
2. It extracts `stx_mnt_id` and iterates upward through directory parents:

$$P_0 = \text{target.parent}(), \quad P_{i+1} = P_i.\text{parent()} \quad (2)$$

3. The traversal terminates at the exact directory P_k where $P_{k+1}.\text{mnt_id} \neq P_k.\text{mnt_id}$ or when reaching filesystem root (`/`).

4. The candidate storage path is evaluated at:

$$\text{StorageCandidate} = P_k / \text{“.rinode-storage”} \quad (3)$$

If the user possesses write permissions on the mount root (common on secondary volumes, external drives, and dedicated home subvolumes), the storage directory is created directly on that mount. If the mount root is read-only or restricted to root (e.g., standard / owned by UID 0), rinode falls back gracefully to the user’s localized state directory (`~/.local/share/rinode/storage`), ensuring the primary home filesystem retains zero-copy preservation.

3.3 Collision-Proof Unique Vault Identifier Generation

When files are moved into a flat vault directory, name collisions are inevitable (e.g., deleting multiple files named `Makefile` from different repositories). rinode resolves this using a structured, deterministic naming scheme:

$$\text{VaultFilename} = \langle \text{Name} \rangle_ \langle \text{Inode} \rangle_ \langle \text{DevMinor} \rangle_ \langle \text{Nanos} \rangle_ \langle \text{Entropy} \rangle \quad (4)$$

Where:

- $\langle \text{Name} \rangle$: Sanitized original filename preserving alphanumerics, dots, and hyphens.
- $\langle \text{Inode} \rangle$: The 64-bit physical inode number retrieved via `stx_ino`.
- $\langle \text{DevMinor} \rangle$: The 32-bit minor device number ensuring differentiation across identical inode indices on different storage targets.
- $\langle \text{Nanos} \rangle$: UTC timestamp at nanosecond resolution from the monotonic clock.
- $\langle \text{Entropy} \rangle$: A 24-bit cryptographically secure pseudorandom hexadecimal string (`0x000000–0xffffffff`) mitigating sub-nanosecond concurrent execution collisions.

An example generated vault path appears as:

```
report.pdf__1982182_0_1789062920171503320_cac889
```

3.4 Metadata Audit Index and SQLite WAL Mode

Audit metadata is stored within an embedded SQLite database (`rinode.db`) placed in `~/.local/share/rinode/`. The database operates strictly under **Write-Ahead Logging (WAL)** mode:

```
1 PRAGMA journal_mode = WAL;
2 PRAGMA synchronous = NORMAL;
3 PRAGMA foreign_keys = ON;
```

WAL mode provides two essential guarantees:

1. **Lock-Free Concurrency**: Readers (such as the interactive terminal user interface) never block writers (CLI `rinode rm` operations), and writers do not block readers.
2. **Sequential I/O**: Modifications are appended to the WAL file sequentially with normal `fsync` semantics, minimizing disk head repositioning.

The database schema captures complete provenance:

```

1 CREATE TABLE IF NOT EXISTS entries (
2     id INTEGER PRIMARY KEY AUTOINCREMENT,
3     filename TEXT NOT NULL,
4     original_path TEXT NOT NULL,
5     vault_path TEXT NOT NULL,
6     inode_no INTEGER NOT NULL,
7     dev_major INTEGER NOT NULL,
8     dev_minor INTEGER NOT NULL,
9     mnt_id INTEGER NOT NULL,
10    file_size INTEGER NOT NULL,
11    mode INTEGER NOT NULL,
12    uid INTEGER NOT NULL,
13    gid INTEGER NOT NULL,
14    is_directory BOOLEAN NOT NULL,
15    link_type TEXT NOT NULL,
16    symlink_target TEXT,
17    quick_fingerprint TEXT,
18    status TEXT NOT NULL DEFAULT 'PRESERVED',
19    deleted_at TEXT NOT NULL,
20    restored_at TEXT
21 );
22 CREATE INDEX IF NOT EXISTS idx_entries_status ON entries(status);
23 CREATE INDEX IF NOT EXISTS idx_entries_inode ON entries(inode_no, dev_major,
    dev_minor);
24 CREATE INDEX IF NOT EXISTS idx_entries_filename ON entries(filename);
25 CREATE INDEX IF NOT EXISTS idx_entries_restored_at ON entries(restored_at);

```

3.5 Dual Restoration Semantics: Consume vs. Snapshot Fork

When undeleting a file, two distinct usage patterns emerge, requiring differentiated VFS semantics:

3.5.1 Mode 1: Consume Restoration (Default)

The entry is moved from the vault back to its original path via `renameat2(2)`. Missing parent directories are reconstructed automatically (`mkdir -p`). The vault file dentry is consumed, and the database marks the record status as `RESTORED` with an exact UTC timestamp. This operation takes $O(1)$ time, preserves the original inode number without modification, and releases zero storage blocks.

3.5.2 Mode 2: Snapshot Fork Restoration (`--keep-copy`)

If the user desires to restore a file while retaining an untouched snapshot in storage, the inode cannot simply be moved back. To prevent duplicating blocks on modern filesystems, `rinode` issues an `ioctl(FICLONE)` system call:

```
1 ioctl(destination_fd, FICLONE, source_fd);
```

On `Btrfs` and `XFS`, this instructs the filesystem extent allocator to point the new inode's extent records to the existing physical blocks on disk. The operation completes in microseconds with zero block copies. If invoked on filesystems lacking CoW support (such as standard `ext4`), `rinode` falls back cleanly to kernel-level data streaming via `copy_file_range(2)` or localized buffered copying. The CLI also retains `--keep-vault` as a backward-compatible alias.

4 Syscall-Level Walkthrough & Transactions

4.1 Preservation Transaction Flow

Figure 3 illustrates the step-by-step kernel and database sequence executed during a file deletion:

User executes: `rinode rm /home/user/workspace/report.pdf`

```
-----
1. Exclusion Check    : Evaluates regex rules and prefix path exclusions
2. Metadata Probe    : statx(AT_FDCWD, path, AT_SYMLINK_NOFOLLOW,
                        STATX_ALL | STATX_MNT_ID, &stx)
3. Fingerprint (64KB): Reads first 64KB -> xxHash64 (in-memory)
4. Storage Discovery : Iterative statx parent traversal -> .rinode-storage
5. Unique Identifier : Generates "report.pdf__1982182_0_178906..._cac889"
6. Atomic Move       : renameat2(AT_FDCWD, path, AT_FDCWD, vault_path, 0)
7. Audit Indexing    : BEGIN IMMEDIATE;
                        INSERT INTO entries (...) VALUES (...);
                        COMMIT;
-----
```

Result: Inode 1982182 remains intact; directory dentry updated in O(1) time.

Figure 3: System call execution pipeline for file preservation.

4.2 Restoration Transaction Flow

When restoring via `rinode restore 42`:

1. **Index Query:** The database retrieves the record matching ID 42.
2. **Hierarchy Reconstruction:** The target path `/home/user/workspace/report.pdf` is evaluated. If intermediate directories (`/home/user/workspace/`) have been removed in the interim, `rinode` executes `mkdir(2)` iteratively with mode `0755` to restore the hierarchy.
3. **Collision Detection:** If the destination path already exists on disk:
 - If `--force` is omitted, restoration halts immediately with `EEXIST`.
 - If `--force` is specified, the obstructing dentry is unlinked first.
4. **VFS Relocation:**
 - Under consume mode: `renameat2(AT_FDCWD, vault_path, AT_FDCWD, orig_path, 0)`.
 - Under snapshot fork mode: open destination with `O_CREAT | O_WRONLY`, invoke `ioctl(FICLONE, vault_fd)`.
5. **Audit State Mutation:** The record in SQLite updates `status = 'RESTORED'` and `restored_at = Utc::now()`.

4.3 Space Reclaim & Garbage Collection

To prevent unbounded storage consumption, `rinode` incorporates an automated purge mechanism:

```
1 SELECT id, vault_path, is_directory FROM entries
2 WHERE status = 'PRESERVED'
3 AND deleted_at < datetime('now', '-14 days');
```

For each identified candidate, `rinode` invokes `unlink(vault_path)` for regular files or `remove_dir_all` for directory hierarchies, followed by setting `status = 'PURGED'`. This releases underlying filesystem extent trees only after the configured retention window expires.

5 Edge Cases & Safety Engineering

5.1 Arbitrary Directory Hierarchies

A major advantage of `renameat2` over hardlinks is its treatment of directory structures. While POSIX forbids hardlinking directories, `renameat2` treats a directory dentry identically to a file dentry. Moving a directory containing 500,000 files into `.rinode-storage` updates only the root dentry in the parent directory. The entire subtree—with all constituent inodes, subdirectories, and extent trees—is preserved instantaneously in a single atomic kernel transaction without recursive scanning.

5.2 Symbolic Links

Symbolic links pose unique security and operational challenges:

- Resolving symlinks during deletion can inadvertently vault the referenced target rather than the link dentry itself.
- `rinode` enforces `AT_SYMLINK_NOFOLLOW` across all `statx` and `renameat2` calls.
- For audit integrity, `rinode` reads the literal link target string via `readlink(2)` and stores it in the `symlink_target` metadata column.

5.3 Open File Descriptors and Concurrent Writing

A subtle edge case occurs if an active process maintains an open file descriptor to a file while `rinode rm` is invoked:

1. The process continues writing through its file descriptor (`struct file`).
2. Because `renameat2` moved the inode rather than severing it, writes continue to hit the physical blocks attached to the vaulted inode.
3. When the active process terminates or invokes `close(fd)`, the data on disk reflects all writes performed up to the moment of closure.

This contrasts sharply with standard `rm`, where writes to an unlinked, open file persist in memory or unlinked blocks, vanishing into free block space upon process termination.

5.4 TOCTOU Race Mitigation with File Descriptors

To prevent Time-of-Check to Time-of-Use (TOCTOU) race conditions—where a path is swapped between the moment `statx` inspects it and `renameat2` executes—all internal VFS primitives support relative descriptor operations (`*at` family). Operations can bind directly to parent directory file descriptors using `openat2(2)` with `RESOLVE_BENEATH` or `RESOLVE_NO_SYMLINKS`, ensuring directory traversal remains confined to verified boundaries.

6 Performance & Empirical Complexity Analysis

6.1 Theoretical Complexity

Table 2 defines the asymptotic algorithmic complexity of `rinode` operations:

Table 2: Algorithmic Complexity of Operations

Operation	Time Complexity	Space Complexity	Disk Writes
Preserve Single File	$O(1)$	$O(1)$	1 dentry update + WAL log
Preserve Directory (N files)	$O(1)$	$O(1)$	1 dentry update + WAL log
Restore (Consume)	$O(1)$	$O(1)$	1 dentry update + WAL log
Restore (Reflink CoW)	$O(1)$	$O(1)$	Metadata extent ref update
Restore (ext4 Fallback Copy)	$O(\text{Size})$	$O(1)$	Full payload copy
Purge / Space Reclamation	$O(\text{Extents})$	$O(1)$	Bitmap free

6.2 Empirical Benchmarks

Empirical evaluations were conducted on an NVMe-backed Linux 6.8 kernel system formatted with ext4 and Btrfs filesystems. Tests compared `rinode rm`, standard GNU `rm -rf`, and `gio trash` across three operational profiles:

- **Profile A:** A single 10 GB continuous virtual machine disk file.
- **Profile B:** A nested Linux kernel source tree containing 78,412 files and directories (totaling 3.8 GB).
- **Profile C:** A 50 GB directory spanning secondary mount points.

Table 3: Execution Latency Comparison Across Deletion Operations

Workload	GNU <code>rm</code>	<code>gio trash</code>	<code>rinode rm</code>
10 GB Single File (ext4)	18.2 ms	42.1 ms	0.31 ms
Kernel Source Tree (78k files)	412.5 ms	894.2 ms	0.48 ms
50 GB Cross-Mount Directory	38.4 ms	148,210.0 ms (Copy)	0.36 ms (Local Vault)

As demonstrated in Table 3, `rinode rm` maintains sub-millisecond execution regardless of payload size or tree depth. While GNU `rm` scales with the number of extents and directory inodes that must be freed, and `gio trash` degrades by orders of magnitude when forced to copy cross-mount datasets, `rinode` executes a single atomic dentry pivot.

7 Discussion, Lessons Learned & Future Work

7.1 Kernel Evolution: From `stat` to `statx`

A primary lesson encountered during development was the inadequacy of classic POSIX interfaces for modern multi-volume systems. Classic `stat(2)` and `statvfs(3)` conflate device numbers (`st_dev`) across Btrfs subvolumes, leading naive tools into fatal EXDEV aborts. The adoption of `statx(2)` with `STATX_MNT_ID` proved to be the foundational enabler of deterministic, race-free mount boundary tracking.

7.2 Transparent Deletion Interception

While shell aliasing (`alias rm="rinode rm"`) provides immediate usability, true transparent protection across all processes requires kernel-level hooks:

1. **LD_PRELOAD Hooking:** Intercepting `unlink`, `unlinkat`, and `rmdir` at the C-library layer. While lightweight, this approach fails against statically compiled binaries (e.g., Go/Rust binaries issuing direct `syscall` instructions).

2. **eBPF (Extended Berkeley Packet Filter)**: Utilizing `LSM_PROBE(file_unlink)` or tracepoints to capture unlinks kernel-wide. However, eBPF programs cannot block and synchronously mutate a deletion into a `renameat2` call within user namespace.
3. **FUSE Overlays**: Implementing a transparent virtual filesystem layer that translates unlinks to vault relocations on the fly.

7.3 Network and Pseudo Filesystems

Network storage (NFS, SMB) and pseudo filesystems (`/proc`, `/sys`) exhibit non-standard VFS semantics. On NFS, client-side unlinks often result in “silly renaming” (`.nfsXXXX` artifacts) to preserve open handles. `rinode`’s exclusion engine explicitly skips pseudo filesystems and provides pattern-based filtering for remote network shares.

8 Conclusion

The standard Linux unlink paradigm permanently destroys data extents as a side effect of directory pruning. `rinode` demonstrates that zero-copy, constant-time deleted file preservation is both viable and computationally superior to existing trash and snapshot mechanisms. By operating directly at the intersection of modern Linux VFS primitives—leveraging `statx(2)` for mount identification, `renameat2(2)` for atomic relocation, Write-Ahead Logged SQLite for audit provenance, and `ioctl(FICLONE)` for reflink snapshotting—`rinode` establishes a robust, auditable, and instantaneous undelete architecture for the Linux operating system.

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